RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

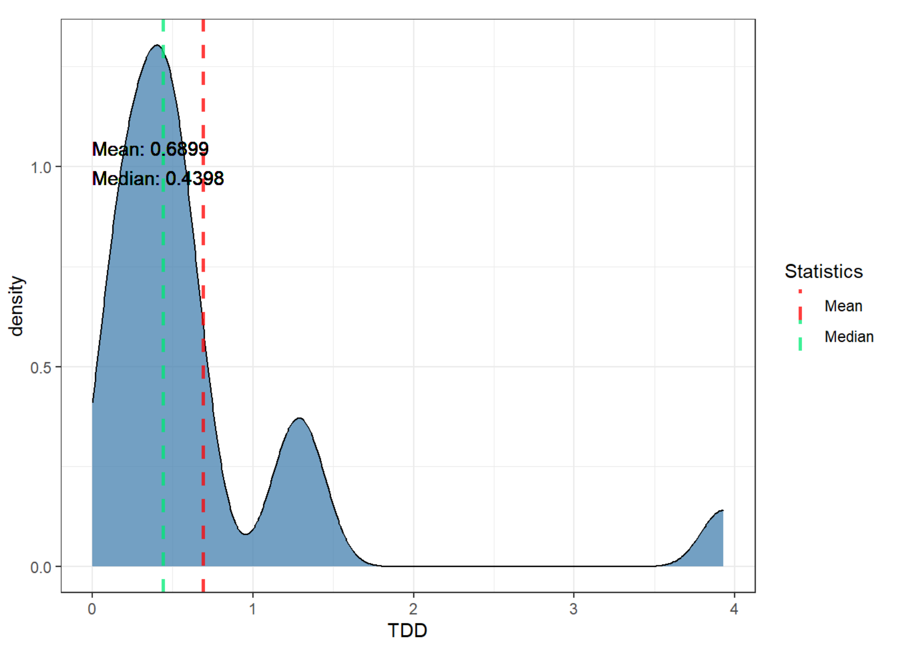
```
## Rows: 1,442
## Columns: 11
## $ repo               <chr> "maven-surefire", "maven-surefire", "maven-surefir…
## $ pr_number          <int> 396, 396, 396, 396, 396, 396, 396, 396, 396, 396, …
## $ rule               <chr> "java:S1192", "java:S1192", "java:S1192", "java:S1…
## $ severity           <chr> "CRITICAL", "CRITICAL", "CRITICAL", "CRITICAL", "C…
## $ file               <chr> "maven-surefire-common/src/test/java/org/apache/ma…
## $ type               <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status             <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "CLOSED", …
## $ debt               <int> 12, 18, 8, 12, 10, 12, 12, 1, 10, 90, 12, 1, 5, 10…
## $ ncloc_affected_file <int> 286, 286, 286, 2015, 2015, 2015, 2015, 2015, 2015,…
## $ complexity         <int> 14, 14, 14, 175, 175, 175, 175, 175, 175, 175, 175…
## $ origin             <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "N…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

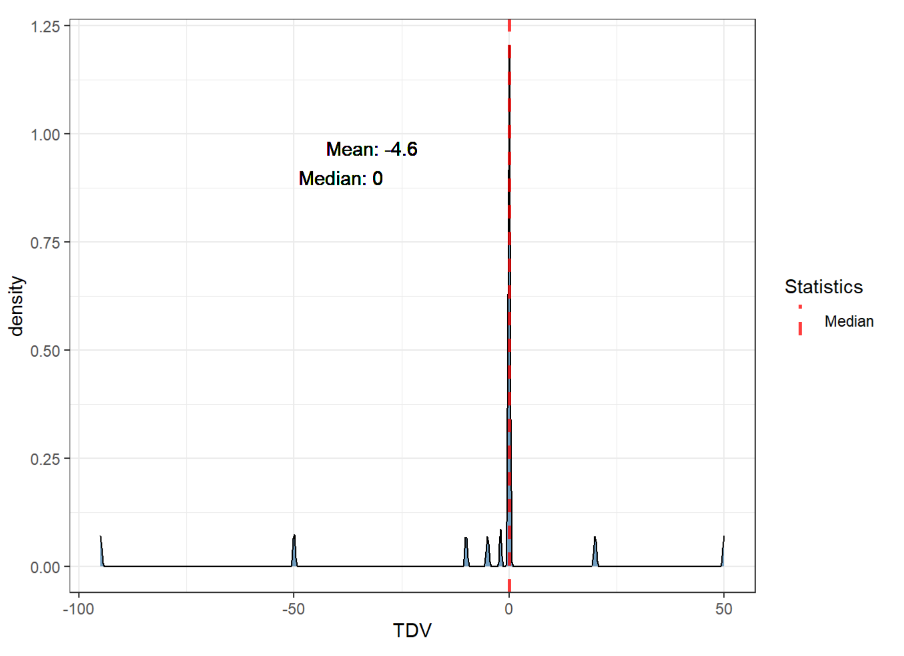


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo           tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                        <dbl>             <dbl>                 <dbl>
## 1 maven-surefire                  25                65                    10
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                       25                     65                         10
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo           preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                                  <dbl>                             <dbl>
## 1 maven-surefire                            65                                35
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                            65                                     35
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo           new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                          <dbl>                     <dbl>
## 1 maven-surefire                   100                         0
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                        100                              0
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 7 × 2
## # Groups:   rank [7]
##    rank `maven-surefire`
##   <int> <chr>
## 1     1 java:S1192  -  5
## 2     2 java:S1874  -  5
## 3     3 java:S112  -  2
## 4     4 java:S1905  -  2
## 5     5 java:S1117  -  1
## 6     6 java:S106  -  1
## 7     7 java:S1075  -  1
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 7 × 2
##   rule       position_score
##   <chr>               <dbl>
## 1 java:S1192            122
## 2 java:S1874            123
## 3 java:S112             124
## 4 java:S1905            125
## 5 java:S106             126
## 6 java:S1075            127
## 7 java:S1117            128
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `maven-surefire`
##    <int> <chr>
##  1     1 java:S1192  -  607
##  2     2 java:S1186  -  180
##  3     3 java:S1874  -  145
##  4     4 java:S106  -  49
##  5     5 java:S3740  -  48
##  6     6 java:S112  -  48
##  7     7 java:S1117  -  34
##  8     8 java:S2696  -  20
##  9     9 java:S1135  -  20
## 10    10 java:S4042  -  18
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192            122
##  2 java:S1186            123
##  3 java:S1874            124
##  4 java:S106             125
##  5 java:S112             126
##  6 java:S3740            127
##  7 java:S1117            128
##  8 java:S1135            129
##  9 java:S2696            130
## 10 java:S4042            131
```

Outliers

Higher TDD

```
## # A tibble: 20 × 5
##    repo           pr_number total_debt total_ncloc   tdd
##    <chr>              <int>      <int>       <int> <dbl>
##  1 maven-surefire       596       1246         317 3.93
##  2 maven-surefire       545         55          40 1.38
##  3 maven-surefire       555        668         519 1.29
##  4 maven-surefire       497        488         409 1.19
##  5 maven-surefire       396       1554        2301 0.675
##  6 maven-surefire       468        322         512 0.629
##  7 maven-surefire       669       1608        2748 0.585
##  8 maven-surefire       619       8474       16261 0.521
##  9 maven-surefire       515        553        1142 0.484
## 10 maven-surefire       608        185         403 0.459
## 11 maven-surefire       435        180         428 0.421
## 12 maven-surefire       508        316         772 0.409
## 13 maven-surefire       527        116         284 0.408
## 14 maven-surefire       507        135         408 0.331
## 15 maven-surefire       668        122         470 0.260
## 16 maven-surefire       454        218         845 0.258
## 17 maven-surefire       444         67         310 0.216
## 18 maven-surefire       412         38         282 0.135
## 19 maven-surefire       564         81         683 0.119
## 20 maven-surefire       490         55         537 0.102
```

Higher TDV

```
## # A tibble: 20 × 3
##    pr_number repo             tdv
##        <int> <chr>          <int>
##  1       545 maven-surefire    50
##  2       668 maven-surefire    20
##  3       412 maven-surefire     0
##  4       444 maven-surefire     0
##  5       490 maven-surefire     0
##  6       497 maven-surefire     0
##  7       507 maven-surefire     0
##  8       508 maven-surefire     0
##  9       527 maven-surefire     0
## 10       555 maven-surefire     0
## 11       564 maven-surefire     0
## 12       596 maven-surefire     0
## 13       608 maven-surefire     0
## 14       619 maven-surefire     0
## 15       669 maven-surefire     0
## 16       396 maven-surefire    -2
## 17       454 maven-surefire    -5
## 18       468 maven-surefire   -10
## 19       515 maven-surefire   -50
## 20       435 maven-surefire   -95
```

Lower TDV

```
## # A tibble: 20 × 3
##    pr_number repo             tdv
##        <int> <chr>          <int>
##  1       435 maven-surefire   -95
##  2       515 maven-surefire   -50
##  3       468 maven-surefire   -10
##  4       454 maven-surefire    -5
##  5       396 maven-surefire    -2
##  6       412 maven-surefire     0
##  7       444 maven-surefire     0
##  8       490 maven-surefire     0
##  9       497 maven-surefire     0
## 10       507 maven-surefire     0
## 11       508 maven-surefire     0
## 12       527 maven-surefire     0
## 13       555 maven-surefire     0
## 14       564 maven-surefire     0
## 15       596 maven-surefire     0
## 16       608 maven-surefire     0
## 17       619 maven-surefire     0
## 18       669 maven-surefire     0
## 19       668 maven-surefire    20
## 20       545 maven-surefire    50
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 20
```

Number of java:S1192 issues that affect test code

```
##     n
## 1 611
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 1
```

Summary NCLOC

```
##      ncloc
##  Min.   :   20.0
##  1st Qu.:  315.2
##  Median :  491.0
##  Mean   : 1373.4
##  3rd Qu.:  790.2
##  Max.   :14200.0
```